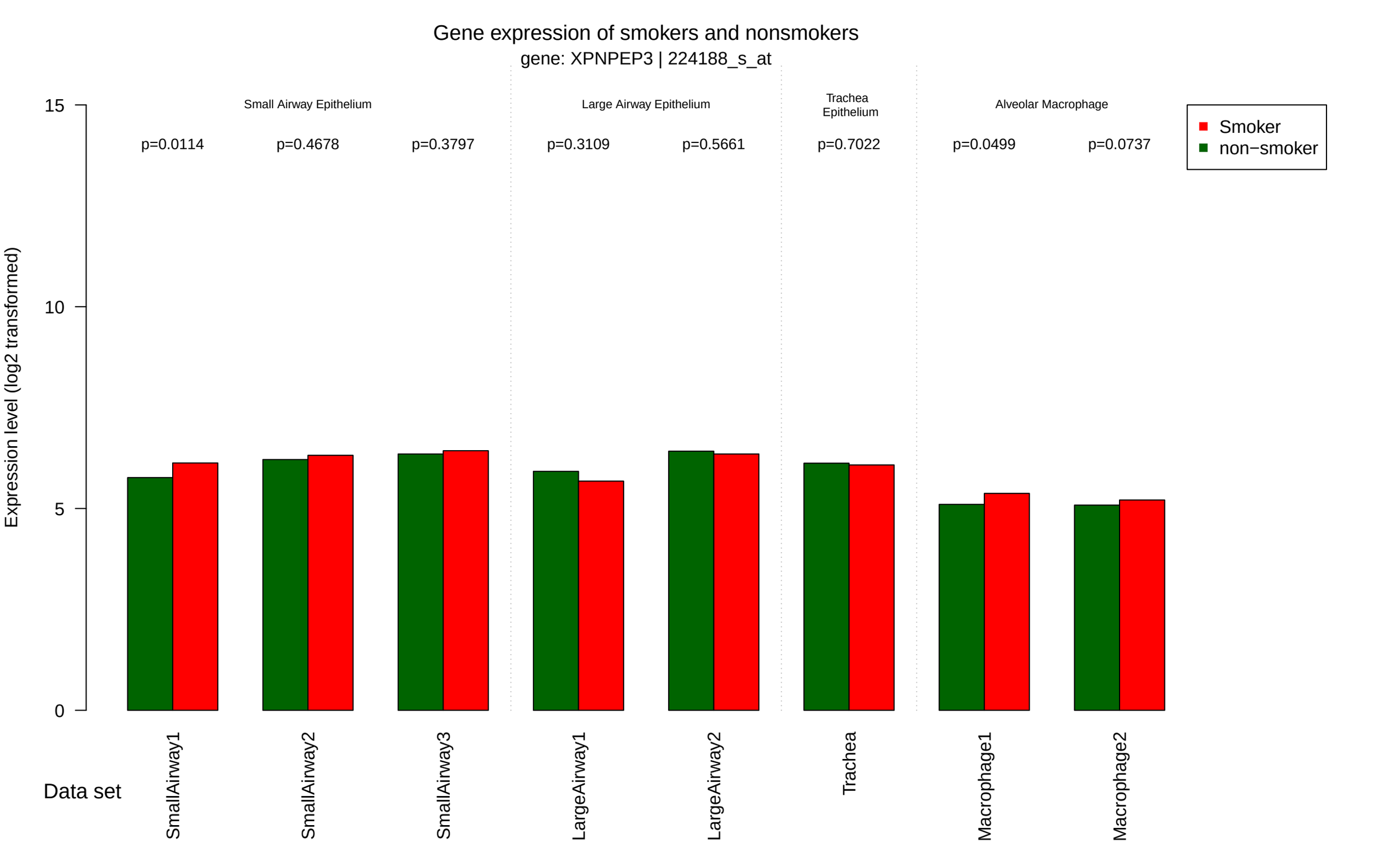
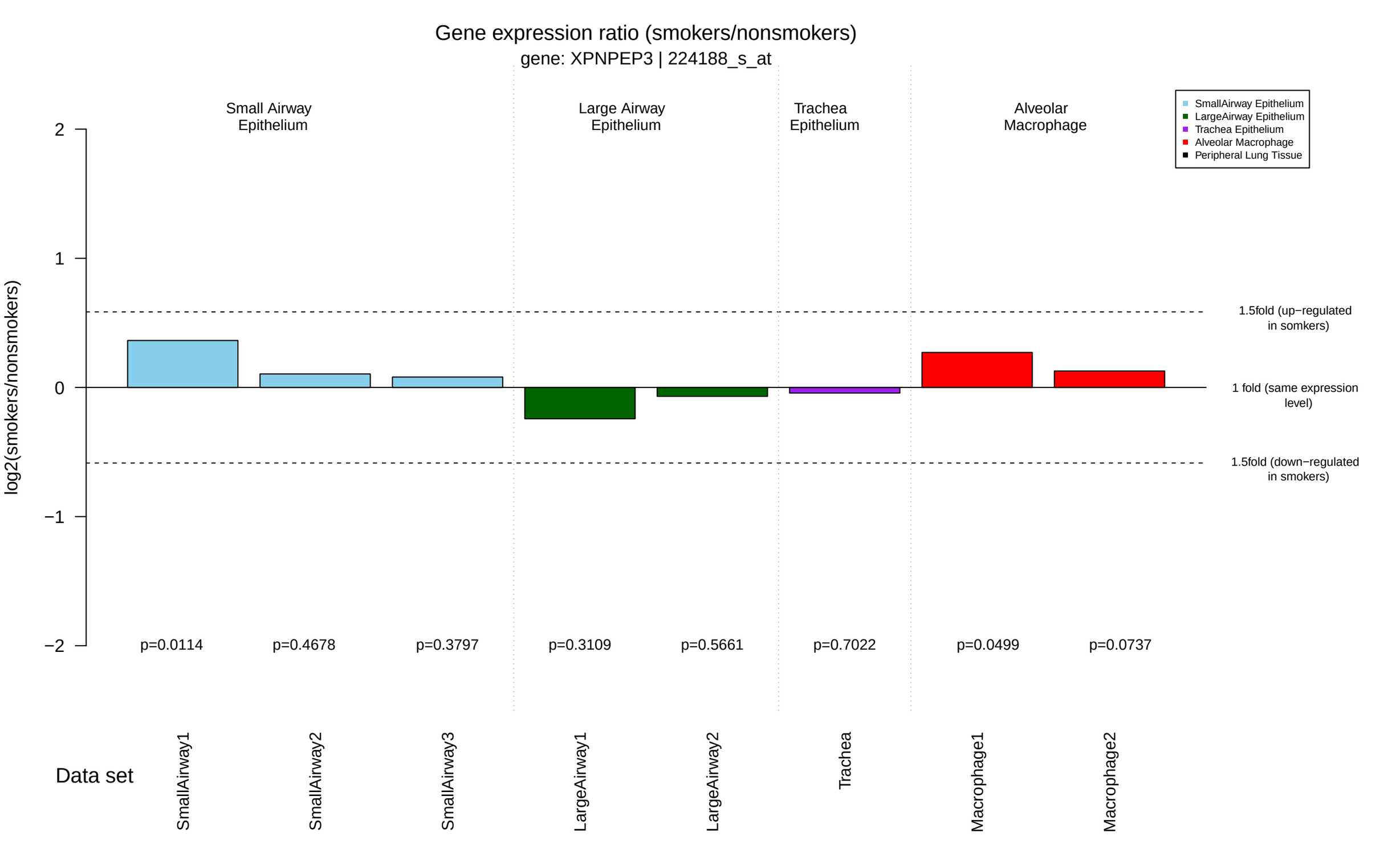
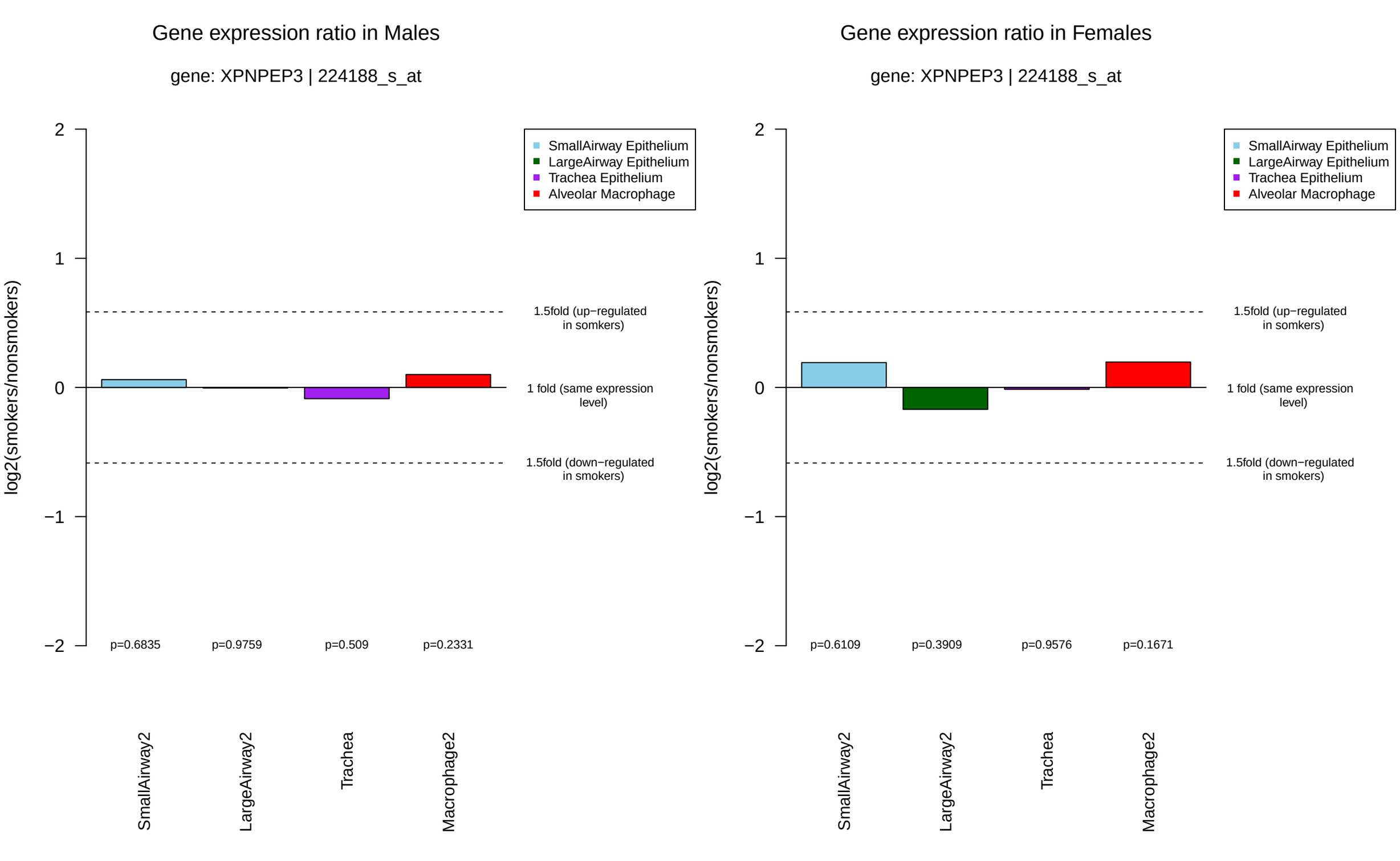
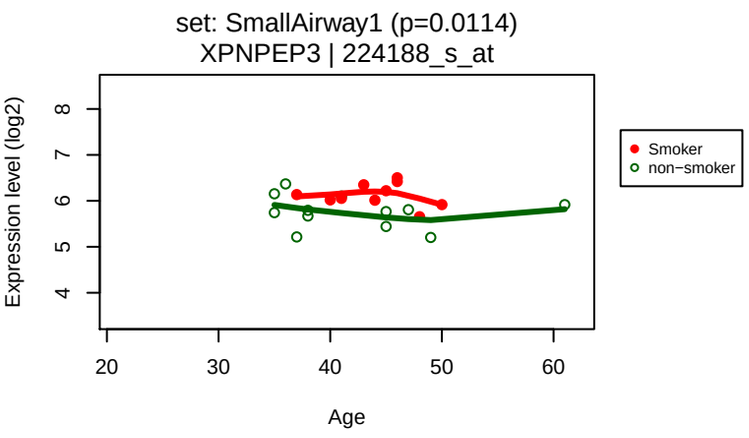
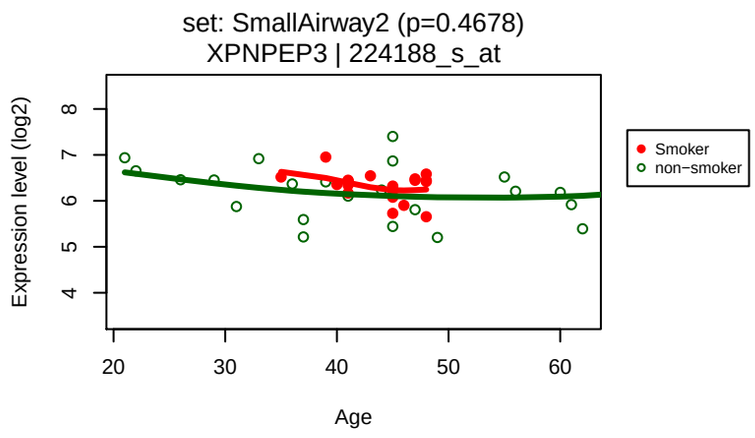
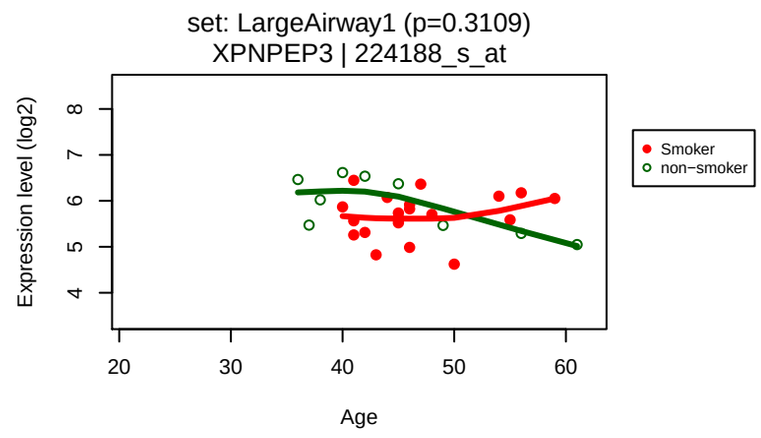

Gene expression of smokers and nonsmokers
gene: XPNPEP3 | 224188_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0114
p=0.4678
p=0.3797
p=0.3109
p=0.5661
p=0.7022
p=0.0499
p=0.0737
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

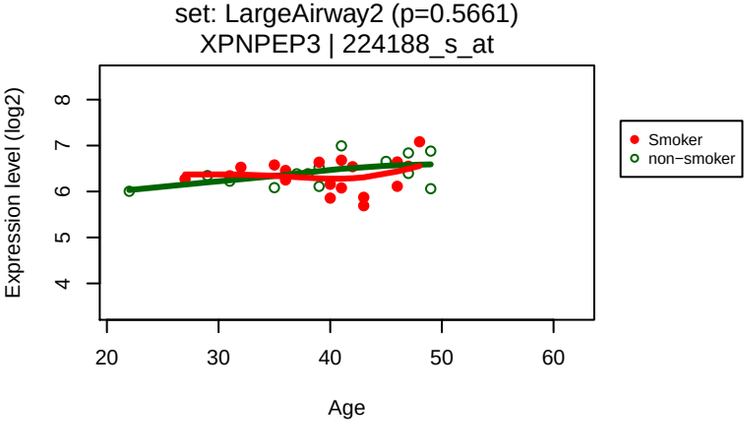

Gene expression ratio (smokers/nonsmokers)
gene: XPNPEP3 | 224188_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0114
p=0.4678
p=0.3797
p=0.3109
p=0.5661
p=0.7022
p=0.0499
p=0.0737
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

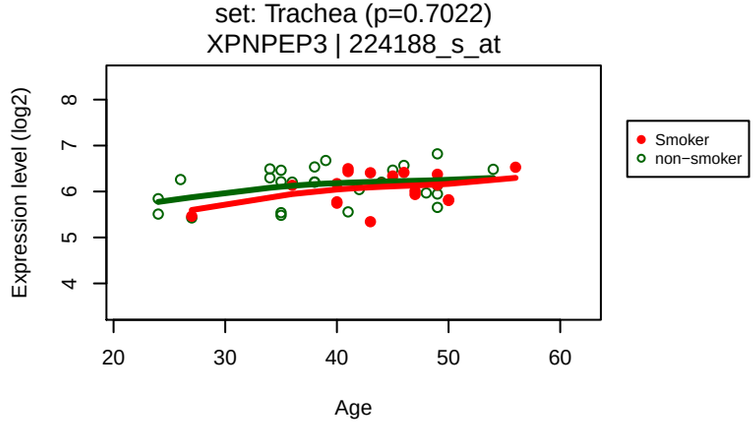

Gene expression ratio in Males
gene: XPNPEP3 | 224188_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6835
p=0.9759
p=0.509
p=0.2331
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: XPNPEP3 | 224188_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6109
p=0.3909
p=0.9576
p=0.1671
SmallAirway2
LargeAirway2
Trachea
Macrophage2

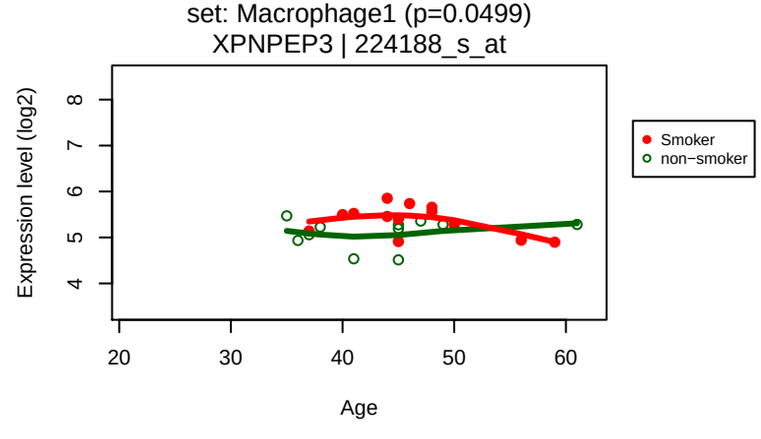

set: SmallAirway1 (p=0.0114)
XPNPEP3 | 224188_s_at
Expression level (log2)
Age
Smoker
non−smoker

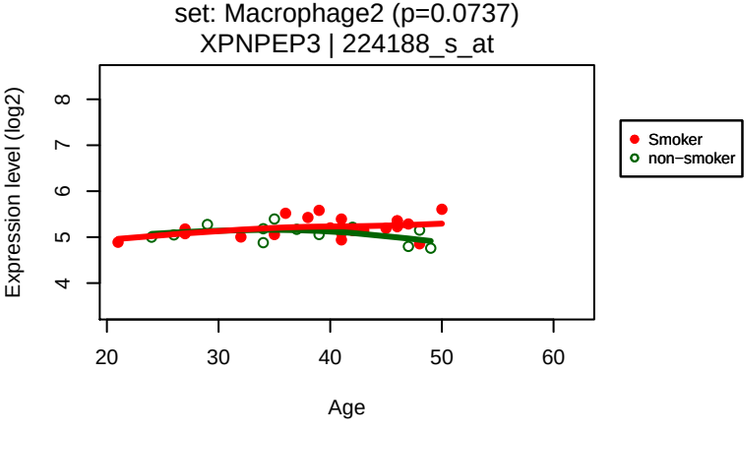

set: SmallAirway2 (p=0.4678)
XPNPEP3 | 224188_s_at
Expression level (log2)
Age
Smoker
non−smoker

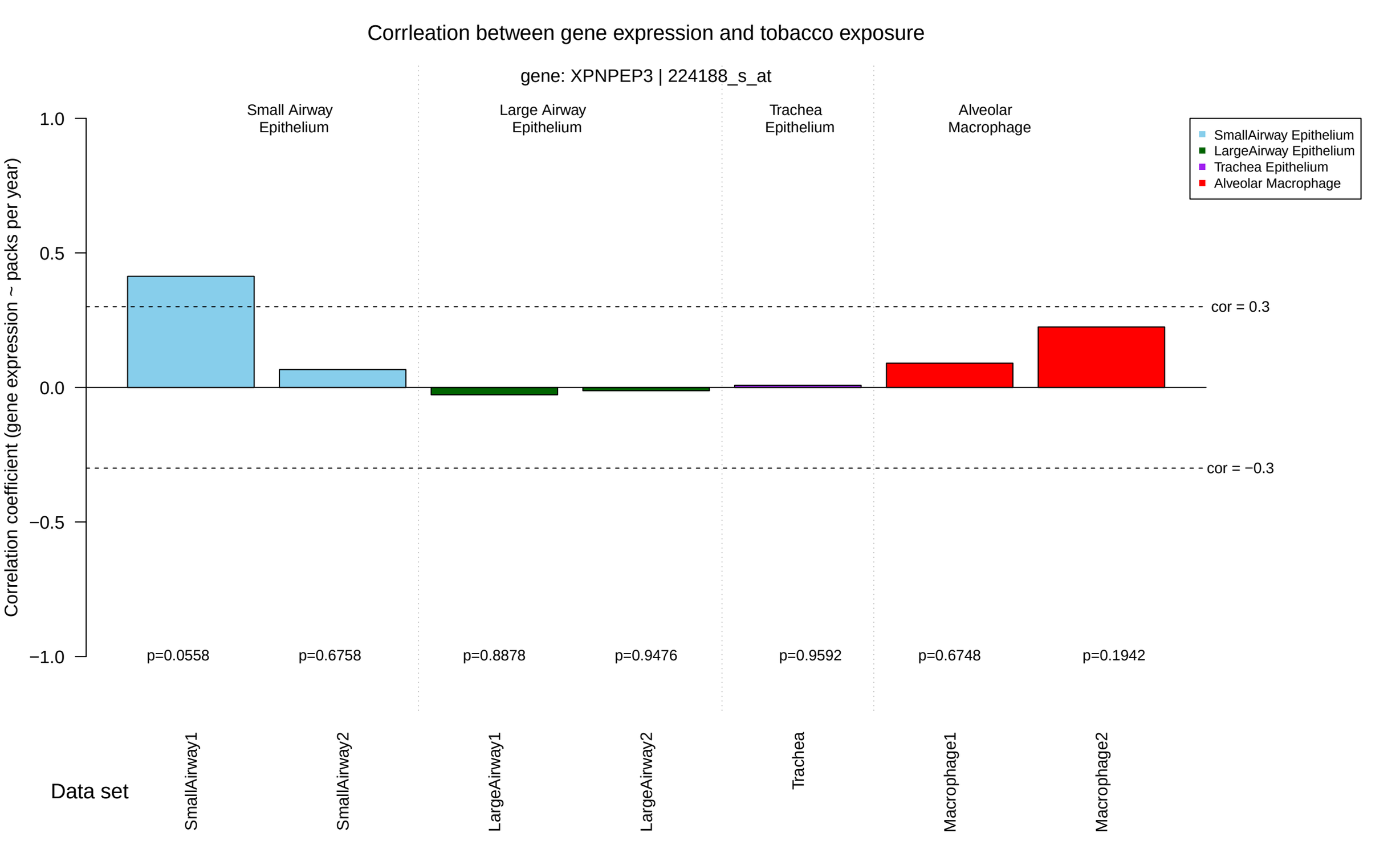

set: LargeAirway1 (p=0.3109)
XPNPEP3 | 224188_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5661)
XPNPEP3 | 224188_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7022)
XPNPEP3 | 224188_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0499)
XPNPEP3 | 224188_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0737)
XPNPEP3 | 224188_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: XPNPEP3 | 224188_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0558
p=0.6758
p=0.8878
p=0.9476
p=0.9592
p=0.6748
p=0.1942
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2